

STAT 5600

Methods in Statistical Learning

FORMAT	CATEGORY	CREDITS
On-Campus	Elective	3
SECTION	BEST ROLE MATCH	AVG ALIGNMENT
Section 4 – Elective (On-Campus)	Machine Learning Engineer (67.6 / 100)	58.9 / 100

59

CURRENT AVG SCORE
across 6 job roles

96

POTENTIAL SCORE
with recommended additions

+44

MAX SCORE GAIN
best single-role improvement

CURRENT COURSE DESCRIPTION

Provides an introduction to methods in the field of statistical learning. Topics include a review of multiple regression, assessing model accuracy, classification, resampling methods, model selection and regularization, nonlinear regression, tree-based methods, support vector machines and unsupervised learning. Involves hands-on data analysis using the R programming language.

INDUSTRY ALIGNMENT BY JOB ROLE

JOB ROLE	CURRENT SCORE	PROJECTED SCORE	POINT GAIN	PRIORITY
Machine Learning Engineer	67.6	100.0	+32.4	HIGH
Data Scientist	66.7	100.0	+33.3	HIGH
AI Engineer	60.9	96.1	+35.2	HIGH
Applied Scientist	59.9	100.0	+40.1	HIGH
ML Engineer	52.0	88.8	+36.8	HIGH
MLOps Engineer	46.2	90.5	+44.2	HIGH

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold (example) appear as gaps across 3 or more job roles.

System Design 4 of 6 roles

systems

Deep Learning 1 of 6 roles

training

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

1 **HIGH** Add System Design coverage — required vocabulary in 4 of 6 target roles. Integrate through lectures, assignments, or lab exercises. Key terms: systems.

2 **LOW** Optional: Deep Learning — would broaden appeal to one additional role. Candidate terms: training.

GAP KEYWORDS BY ROLE

Machine Learning Engineer

67.6 → 100.0 (+32.4 pts)

predictive tools exploratory design development science models applications

Data Scientist

66.7 → 100.0 (+33.3 pts)

toolkit product simply different social problems approaches

AI Engineer

60.9 → 96.1 (+35.2 pts)

models systems problems detection machine design product tools

Applied Scientist

59.9 → 100.0 (+40.1 pts)

biology computational research problems models systems design real

ML Engineer

52.0 → 88.8 (+36.8 pts)

machine systems models problems product tools detection

MLOps Engineer

46.2 → 90.5 (+44.2 pts)

detection systems machine image training tools models development

Methodology: Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum-Industry Alignment project team.